AUDIT SISTEM INFORMASI

Minggu ke-3

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Pokok Bahasan

- Design Factor
- Impact of Design Factors

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Faktor desain adalah faktor-faktor yang dapat memengaruhi desain sistem tata kelola perusahaan dan memposisikannya untuk sukses dalam penggunaan TIK. Faktor desain tersebut tercantum di bawah ini beserta potensi dampaknya pada sistem tata kelola.



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Setiap sektor industri memiliki serangkaian persyaratan uniknya sendiri terkait harapan dari penggunaan TI. Namun, dimungkinkan untuk menangkap karakteristik utama suatu sektor industri dengan kombinasi desain factor (DF), Misalnya:

- Sektor keuangan dapat dicirikan sebagai berikut: TI sangat diatur, TI memainkan peran strategis, biasanya terdiri dari perusahaan besar, dan beroperasi dalam lingkungan dengan ancaman tinggi.
- Penyedia layanan kesehatan (misalnya, rumah sakit) biasanya bertujuan untuk kombinasi layanan/stabilitas klien dan strategi inovasi, sangat diatur, tunduk pada sejumlah area risiko spesifik (keselamatan, keamanan, privasi, keberlanjutan, dll.), beroperasi dalam lingkungan dengan ancaman moderat (tetapi meningkat), dan semakin bergantung secara strategis pada TI.

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- Lembaga nirlaba biasanya lebih kecil, dan kurang diatur, berfokus pada biaya, dan bukan inovator utama dalam adopsi teknologi.

- Lembaga sektor publik seringkali merupakan organisasi besar, dengan strategi layanan klien dan kepemimpinan biaya. Mereka memiliki profil risiko sedang hingga tinggi dan sangat diatur karena sifatnya. Peran TI dapat bervariasi, dari dukungan di lembaga konservatif, hingga strategis dalam hal inisiatif e-government. Model pengadaan semakin banyak menggunakan layanan yang dialihdayakan, sedangkan mereka umumnya mengikuti arus utama dalam adopsi teknologi.

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C0BIT® 2019 memiliki 11 Desain Factor:

1. **Enterprise strategy**
2. **Enterprise goals**
3. **Risk profile**
4. **I&T-related issues**
5. **Threat landscape**
6. **Compliance requirements**
7. **Role of IT**
8. **Sourcing model for IT**
9. **IT implementation methods**
10. **Technology adoption strategy**
11. **Enterprise size**

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Desain Factor -1 **Enterprise strategy**

Enterprise strategy—Perusahaan dapat memiliki strategi yang berbeda, yang dapat diungkapkan sebagai satu atau lebih dari archetypes yang ditunjukkan pada gambar . Organisasi biasanya memiliki strategi utama dan, paling banyak, satu strategi sekunder.

Strategy Archetype	Explanation
Growth/Acquisition	The enterprise has a focus on growing (revenues)
Innovation/Differentiation	The enterprise has a focus on offering different and/or innovative products and services to their clients
Cost Leadership	The enterprise has a focus on short-term cost minimization
Client Service/Stability	The enterprise has a focus on providing a stable and client-oriented service.

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Desain Factor – 2 **Enterprise goals**

Enterprise goals yang mendukung strategi perusahaan—Strategi perusahaan diwujudkan melalui pencapaian (seperangkat) tujuan perusahaan.

Tujuan-tujuan ini didefinisikan dalam kerangka kerja COBIT. kerangka kerja COBIT disusun berdasarkan dimensi balanced scorecard (BSC), dan mencakup hal-hal berikut:

BSC dimension	Ref.	Enterprise Goal
Financial	EG01	Portfolio of competitive products and services
	EG02	Managed business risk
	EG03	Compliance with external laws and regulations
	EG04	Quality of financial information
Customer	EG05	Customer-oriented service culture
	EG06	Business-service continuity and availability
	EG07	Quality of management information
Internal	EG08	Optimization of internal business process functionality
	EG09	Optimization of business process costs
	EG10	Staff skills, motivation and productivity
	EG11	Compliance with internal policies
Learning and Growth	EG12	Managed digital transformation programs
	EG13	Product and business innovation

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Desain Factor -3 **Risk profile**

Risk profile perusahaan dan isu-isu terkini terkait TI—Profil risiko mengidentifikasi jenis risiko terkait TI yang saat ini dihadapi perusahaan dan menunjukkan area risiko mana yang melebihi toleransi risiko.

COBIT® 2019 memiliki 19 **Risk profile** sebagai berikut:

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Reference	Risk Category	Example Risk Scenarios
1	IT-investment decision making, portfolio definition and maintenance	A. Programs selected for implementation misaligned with corporate strategy and priorities B. Failure of IT-related Investments to support digital strategy of the enterprise C. Selection of wrong software (in terms of cost, performance, features, compatibility, redundancy, etc.) for acquisition and implementation D. Selection of wrong infrastructure (in terms of cost, performance, features, compatibility, etc.) for implementation E. Duplication or important overlaps between different investment initiatives F. Long-term incompatibility between new investment programs and enterprise architecture G. Misallocation, inefficient management and/or competition for resources without alignment to business priorities
2	Program and projects lifecycle management	A. Failure of senior management to terminate failing projects (due to cost explosion, excessive delays, scope creep, changed business priorities) B. Budget overruns for I&T projects C. Lack of quality of I&T projects D. Late delivery of I&T projects E. Failure of third-party outsourcers to deliver projects as per contractual agreements (any combination of exceeded budgets, quality problems, missing functionality, late delivery)
3	IT cost and oversight	A. Extensive dependency on, and use of, user-created, user-defined, user-maintained applications and *ad hoc* solutions B. Excess cost and/or ineffectiveness of I&T-related purchases outside of the I&T procurement process C. Inadequate requirements leading to ineffective Service Level Agreements (SLAs) D. Lack of funds for I&T related investments
4	IT expertise, skills and behavior	A. Lack or mismatch of IT-related skills within IT (e.g., due to new technologies or working methods) B. Lack of business understanding by IT staff that affects service delivery/project quality C. Inability to recruit and retain IT staff D. Recruitment of unsuitable profiles because of lack of due diligence in the recruitment process E. Lack of I&T training F. Overreliance for I&T services on key staff

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Reference	Risk Category	Example Risk Scenarios
5	Enterprise/IT architecture	A. Complex, inflexible enterprise architecture (EA), obstructing further evolution and expansion, and leading to missed business opportunities B. Failure to timely adopt and exploit new infrastructure or abandon obsolete infrastructure C. Failure to timely adopt and exploit new software (functionality, optimization, etc.) or to abandon obsolete applications D. Undocumented EA leading to inefficiencies and duplications E. Excessive number of exceptions on enterprise architecture standards
6	IT operational infrastructure incidents	A. Accidental damaging of IT equipment B. Errors by IT staff (during backup, during upgrades of systems, during maintenance of systems, etc.) C. Incorrect information input by IT staff or system users D. Destruction of data center (sabotage, etc.) by staff E. Theft of device with sensitive data F. Theft of a key infrastructure component G. Erroneous configuration of hardware components H. Intentional tampering with hardware (security devices, etc.) I. Abuse of access rights from prior roles to access IT infrastructure J. Loss of backup media or backups not checked for effectiveness K. Loss of data by cloud provider L. Operational-service interruption by cloud providers
7	Unauthorized actions	A. Tampering with software B. Intentional modification or manipulation of software leading to incorrect data C. Intentional modification or manipulation of software leading to fraudulent actions D. Unintentional modification of software leading to inaccurate results E. Unintentional configuration and change-management errors

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8	Software adoption/ usage problems	A. Nonadoption of new application software by users B. Inefficient use of new software by users
9	Hardware incidents	A. System instability in wake of installing new infrastructure, leading to operational incidents (e.g., BYOD program) B. Inability of systems to handle transaction volumes when user volumes increase C. Inability of systems to handle load when new applications or initiatives are deployed D. Utilities failure (telecom, electricity) E. Hardware failure due to overheating and/or other environmental conditions like humidity F. Damaging of hardware components leading to destruction of data by internal staff G. Loss/disclosure of portable media containing sensitive data (CD, USB-drives, portable disks, etc.) H. Extended resolution time or support delays in case of hardware incidents
10	Software failures	A. Inability to use the software to realize desired outcomes (e.g., failure to make required business model or organizational changes) B. Implementation of immature software (early adopters, bugs, etc.) C. Operational glitches when new software is made operational D. Regular software malfunctioning of critical application software E. Obsolete application software (outdated, poorly documented, expensive to maintain, difficult to extend, not integrated in current architecture, etc.) F. Inability to revert back to former versions in case of operational issues with a new version G. Software-induced corrupted data(base) leading to inaccessible data

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Reference	Risk Category	Example Risk Scenarios
11	Logical attacks (hacking, malware, etc.)	A. Unauthorized (internal) users trying to break into systems B. Service interruption due to denial-of-service (DoS) attack C. Website defacement D. Malware attack E. Industrial espionage F. Hacktivism G. Disgruntled employee implements a time bomb which leads to data loss H. Company data stolen through unauthorized access gained by a phishing attack I. Foreign government attacks on critical systems
12	Third-party/supplier incidents	A. Inadequate performance of outsourcer in large-scale, long-term outsourcing arrangement (e.g., through lack of supplier due diligence regarding financial viability, delivery capability and sustainability of supplier's service) B. Accepting unreasonable terms of business from IT suppliers C. Inadequate support and services delivered by vendors, not in line with SLA D. Noncompliance with software license agreements (use and/or distribution of unlicensed software) E. Inability to transfer to alternative suppliers due to overreliance or overdependence on current supplier F. Purchase of IT services (especially cloud services) by the business without consultation /involvement of IT, resulting in inability to integrate the service with in-house services. G. Inadequate or unenforced SLA to obtain agreed services and penalties in case of noncompliance
13	Noncompliance	A. Noncompliance with national or international regulations (e.g., privacy, accounting, manufacturing, environmental, etc.) B. Lack of awareness of potential regulatory changes that may have a business impact C. Operational obstacles caused by regulations D. Failure to comply with internal procedures

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14	Geopolitical issues	A. Lack of access due to disruptive incident in other premises B. Government interference and national policies impacting the business C. Targeted action from government-sponsored groups or agencies
15	Industrial action	A. Facilities and building inaccessible because of labor union strike B. Third-party providers unable to provide services because of strike C. Key staff unavailable through industrial action (e.g., transportation or utilities strike)
16	Acts of nature	A. Earthquake destroying or damaging important IT infrastructure B. Tsunami destroying critical premises C. Major storms and tropical cyclone or tornado damaging critical infrastructure D. Major wildfire E. Flooding F. Rising water table leaving critical location unusable G. Rising temperature rendering critical locations uneconomical to operate
17	Technology-based innovation	A. Failure to identify new and important technology trends B. Failure to appreciate the value and potential of new technologies C. Failure to adopt and exploit new technologies in a timely manner (functionality, process optimization, etc.) D. Failure to provide technology support new business models
18	Environmental	A. Environmentally unfriendly equipment (e.g., power consumption, packaging)
19	Data and information management	A. Discovery of sensitive information by unauthorized persons due to inefficient retaining/archiving/disposing of information B. Intentional illicit or malicious modification of data C. Unauthorized disclosure of sensitive information through email or social media D. Loss of IP and/or leakage of competitive information

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Desain Factor – 4 **I&T-related issues**

I&T-related issues—Metode terkait untuk penilaian risiko TI bagi perusahaan adalah dengan mempertimbangkan isu-isu terkait TI apa yang saat ini dihadapinya, atau dengan kata lain, risiko terkait TI apa yang telah terwujud.

Isu-isu yang paling umum dalam COBIT® 2019 ada 20 isu terkait TI (ditulis dalam huruf) antara lain:

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Reference	Description
A	Frustration between different IT entities across the organization because of a perception of low contribution to business value
B	Frustration between business departments (i.e., the IT customer) and the IT department because of failed initiatives or a perception of low contribution to business value
C	Significant IT related incidents, such as data loss, security breaches, project failure, application errors, etc. linked to IT
D	Service delivery problems by the IT outsourcer(s)
E	Failures to meet IT related regulatory or contractual requirements
F	Regular audit findings or other assessment reports about poor IT performance or reported IT quality or service problems
G	Substantial hidden and rogue IT spending, that is, IT spending by user departments outside the control of the normal IT investment decision mechanisms and approved budgets
H	Duplications or overlaps between various initiatives or other forms of wasting resources
I	Insufficient IT resources, staff with inadequate skills or staff burnout/dissatisfaction
J	IT-enabled changes or projects frequently failing to meet business needs and delivered late or over budget
K	Reluctance by board members, executives or senior management to engage with IT, or lack of committed business sponsors for IT

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L	Complex IT operating model and/or unclear decision mechanisms for IT-related decisions
M	Excessively high cost of IT
N	Obstructed or failed implementations of new initiatives or innovations caused by the current IT architecture and system
O	Gap between business and technical knowledge which leads to business users and IT and/or technology specialists speaking different languages
P	Regular issues with data quality and integration of data across various sources
Q	High level of end-user computing, creating (among other problems) a lack of oversight and quality control over the applications that are being developed and put in operation
R	Business departments implementing their own information solutions with little or no involvement of the enterprise IT department [8]
S	Ignorance and/or noncompliance with security and privacy regulations
T	Inability to exploit new technologies or to innovate using I&T

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Desain Factor – 5 **Threat landscape**

Threat landscape—Lanskap ancaman yang dihadapi perusahaan dapat diklasifikasikan seperti yang ditunjukkan sebagai berikut

Threat Landscape	Explanation
Normal	The enterprise is operating under what are considered normal threat levels
High	Due to its geopolitical situation, industry sector or particular profile, the enterprise is operating in a high-threat environment.

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Desain Factor – 6 **Compliance requirements**

Compliance requirements—persyaratan kepatuhan yang harus dipatuhi oleh perusahaan dapat diklasifikasikan menurut kategori yang tercantum pada gambar berikut

Regulatory Environment	Explanation
Low compliance requirements	The enterprise is subject to a minimal set of regular compliance requirements that are lower than average.
Normal compliance requirements	The enterprise is subject to a set of regular compliance requirements that are common across different industries.
High compliance requirements	The enterprise is subject to higher than average compliance requirements, most often related to industry sector or geopolitical conditions.

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Desain Factor – 7 **Role of IT**

Role of IT—Peran TI bagi perusahaan dapat diklasifikasikan seperti yang ditunjukkan pada gambar berikut.

Role of IT[9]	Explanation
Support	IT is not crucial for the running and continuity of the business process and services, nor for their innovation.
Factory	When IT fails, there is an immediate impact on the running and continuity of the business processes and services. However, IT is not seen as a driver for innovating business processes and services.
Turnaround	IT is seen as a driver for innovating business processes and services. At this moment, however, there is not a critical dependency of IT for the current running and continuity of the business processes and services.
Strategic	IT is critical for both running and innovating the organization's business processes and services.

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Desain Factor – 8 **Sourcing model for IT**

Sourcing model for IT—Model pengadaan untuk TI yang diadopsi perusahaan dapat diklasifikasikan seperti yang ditunjukkan pada gambar berikut.

Sourcing Model	Explanation
Outsourcing	The enterprise calls upon the services of a third party to provide IT services.
Cloud	The enterprise maximizes the use of the cloud for providing IT services to its users.
Insourced	The enterprise provides for their own IT staff and services.
Hybrid	A mixed model is applied, combining the three models above in varying degrees.

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Desain Factor – 9 IT implementation methods

IT implementation methods —Metode implementasi TI yang diadopsi perusahaan dapat diklasifikasikan seperti yang tercantum pada gambar berikut.

IT Implementation Method	Explanation
Agile	The enterprise uses Agile development working methods for its software development.
DevOps	The enterprise uses DevOps working methods for software building, deployment and operations.
Traditional	The enterprise uses a more classic approach towards software development (waterfall) and separates software development and operations.
Hybrid	The enterprise uses a mix of traditional and modern IT implementation, often referred to as "bimodal IT."

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Desain Factor – 10 **Technology adoption strategy**

Technology adoption strategy—Strategi adopsi teknologi dapat diklasifikasikan seperti yang tercantum pada gambar berikut.

Technology Adoption Strategy	Explanation
First mover	The enterprise generally adopts new technologies as early as possible and tries to gain first-mover advantage.
Follower	The enterprise typically waits for new technology to become mainstream and proven before adopting them.
Slow adopter	The enterprise is very late with their adoption of new technologies.

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Desain Factor – 11 **Enterprise size**

Enterprise size—Dua kategori, seperti yang ditunjukkan pada gambar, diidentifikasi untuk desain sistem tata kelola perusahaan.

Enterprise Size	Explanation
Large enterprise (default)	Enterprises with more than 250 full-time employees (FTEs)
Small and medium enterprise	Enterprise with 50 to 250 FTEs

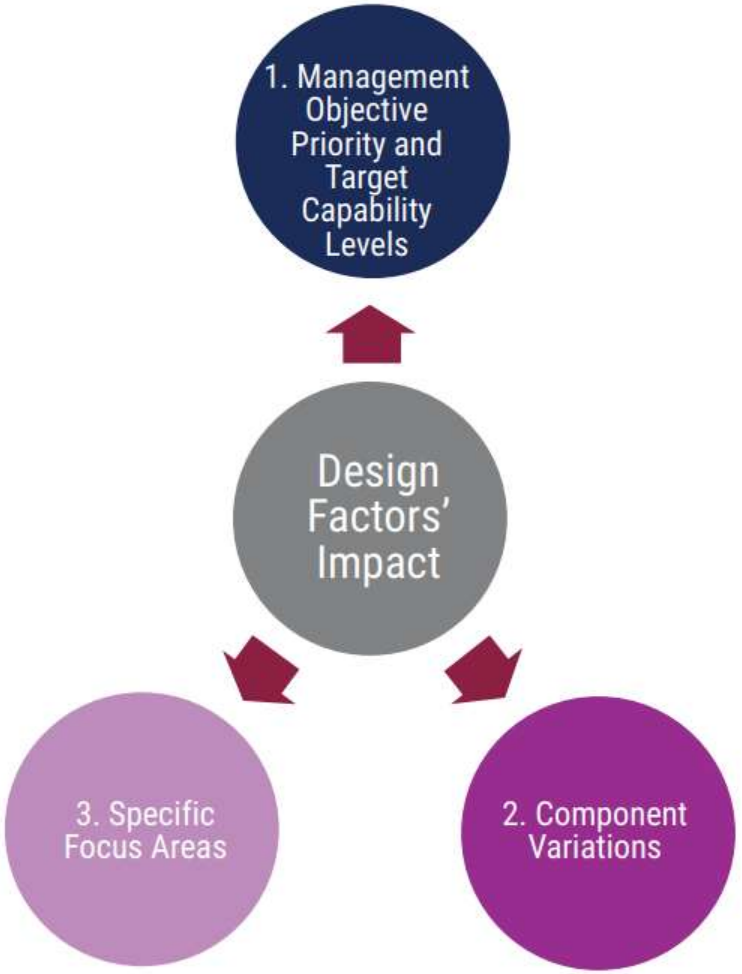
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Impact of Design Factors

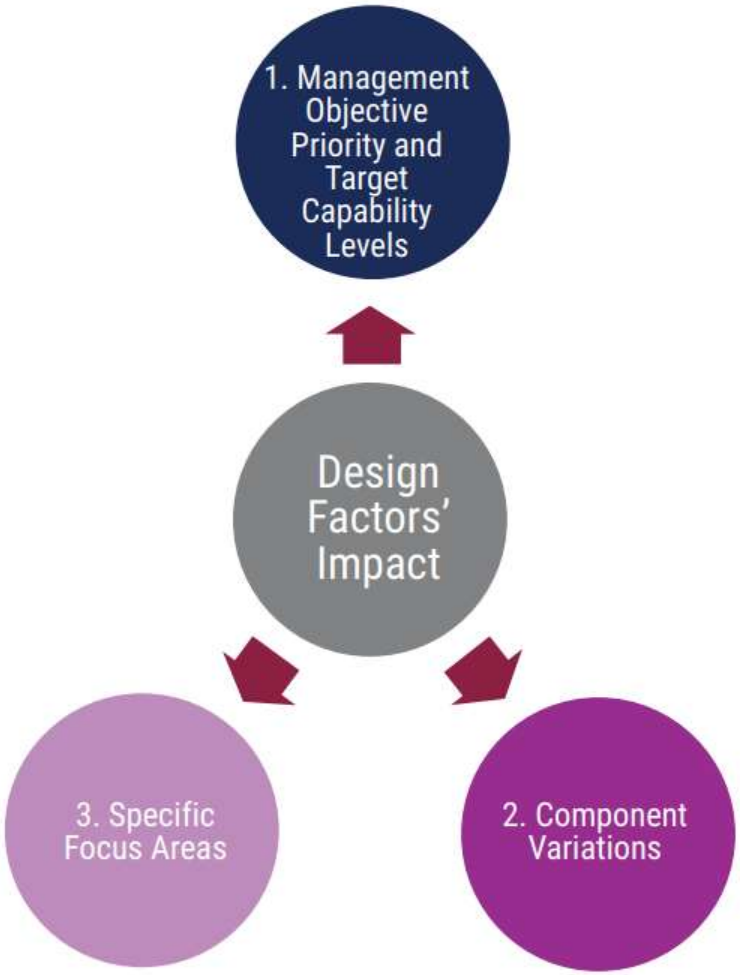
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Dampak dari Desain Faktor akan mempengaruhi 3 hal:

1. Management Objective Priority, Target Capability Level
2. Komponen diperlukan untuk mencapai tujuan Government and Management
3. Fokus area perbaikan/peningkatan



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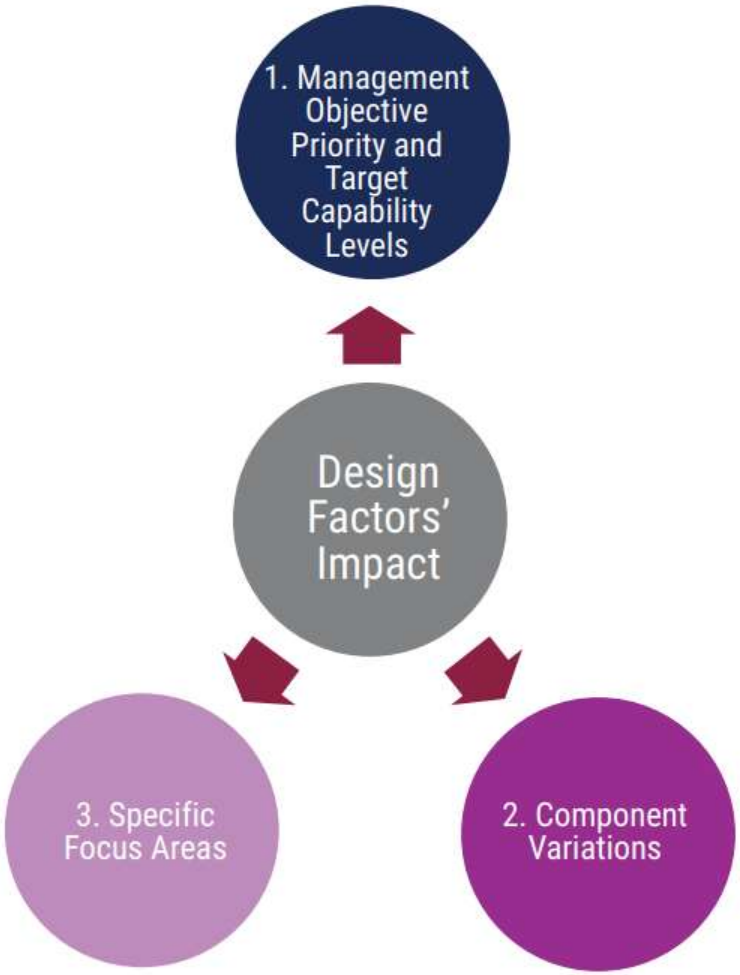
1. Management objective priority/selection —Model inti COBIT berisi 40 tujuan tata kelola dan manajemen, masing-masing terdiri dari proses dan sejumlah komponen terkait.

Tujuan-tujuan ini pada dasarnya setara; tidak ada urutan prioritas alami di antara ke-40 tujuan tata kelola dan manajemen.

Namun, faktor desain dapat memengaruhi kesetaraan ini dan membuat beberapa tujuan tata kelola dan manajemen lebih penting daripada yang lain. Dalam praktiknya, kepentingan/prioritas yang lebih tinggi ini diterjemahkan menjadi target capability levels.

Contoh: Suatu perusahaan yang beroperasi dalam lingkungan dengan ancaman tinggi akan membutuhkan *highly capable security-related processes maka komponen COBIT yang paling relevan diantaranya*: APO13 *Managed security* dan DSS05 *Managed security services* .

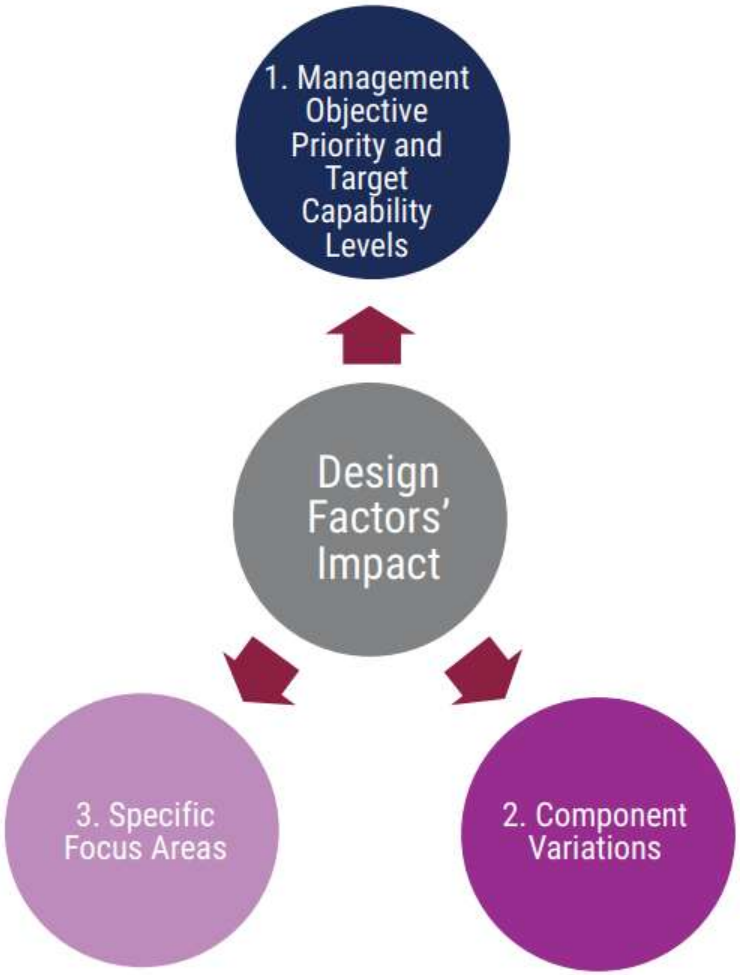
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2. Components variation —Variasi komponen: Komponen diperlukan untuk mencapai tujuan tata kelola dan manajemen. Faktor desain dapat mewajibkan variasi komponen tertentu atau dapat memengaruhi pentingnya komponen.

Contoh: Suatu perusahaan yang beroperasi di lingkungan yang sangat teratur akan lebih berfokus pada produk kerja. Perusahaan ini akan memiliki kebijakan serta prosedur yang terdokumentasi.

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3. Need for specific focus area guidance —Kebutuhan akan panduan area fokus spesifik: beberapa faktor desain, seperti lanskap ancaman, risiko spesifik, metode pengembangan target, pengaturan infrastruktur, akan mendorong kebutuhan akan variasi konten model inti COBIT ke konteks spesifik.

Contoh: Usaha kecil dan menengah yang memiliki jumlah staf yang lebih sedikit, sumber daya TI yang lebih sedikit, dan jalur pelaporan yang lebih pendek dan lebih langsung, dan berbeda dalam banyak aspek dari perusahaan besar. Karena alasan itu, sistem tata kelola TI mereka harus lebih ringan dibandingkan dengan perusahaan besar.

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